What Egocentric Video Embeddings Encode and How to Combine Them with Inertial Sensors for Human Activity Recognition

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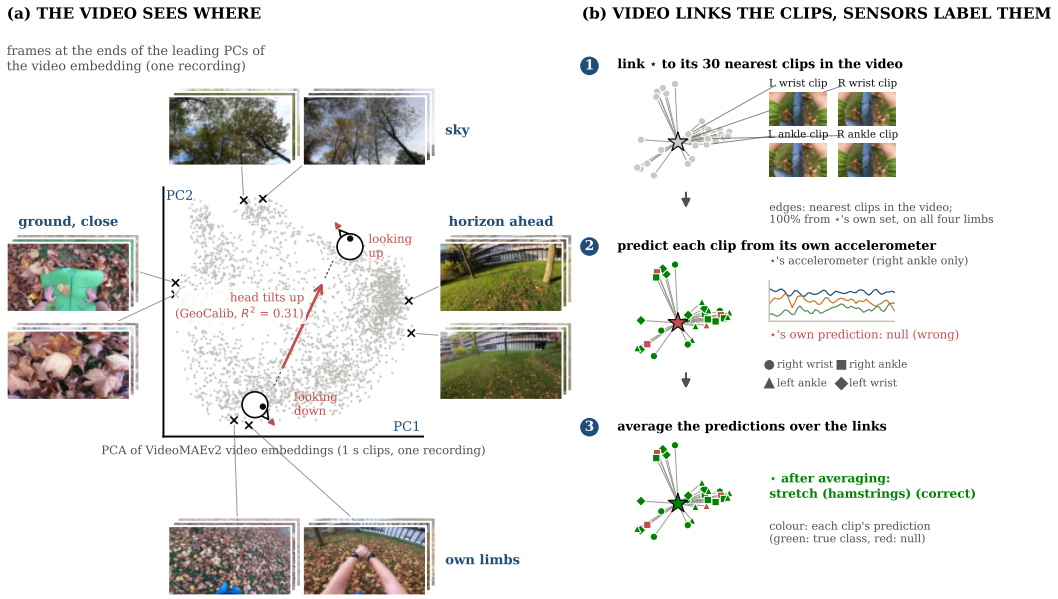


Fig. 1. What egocentric video embeddings encode and how they combine with inertial sensing, on one WEAR recording. (a) Frames at the two ends of the first and second principal components of the video embedding (VideoMAEv2). The components follow where the camera looks. The arrow is the gradient of head tilt estimated from the frames (GeoCalib [7]). Figure 11 shows three participants. (b) The video links the clips and the sensors label them, shown for a held-out clip ⋆ of a training participant (cross-validation). (1) ⋆ is linked to its 30 nearest clips in the video (edges, with one photo per limb); all come from its own set and are spread over the four limbs. (2) Each clip is predicted from its own accelerometer (colour; shape: limb). As at test time, ⋆ carries one limb, and its own prediction is wrong. (3) Averaging the predictions over the links corrects ⋆. Frames from WEAR [1] (CC BY-NC-SA 4.0, https://creativecommons.org/licenses/by-nc-sa/4.0/), downscaled.

Egocentric video and body-worn inertial sensors are increasingly recorded together, and the video is usually encoded by pretrained models. What these embeddings represent when the camera sits on the head, and how they should be combined with inertial sensing, has received little attention. Across VideoMAEv2, CLIP, DINOv2 and the egocentric VC-1, the leading principal directions of the embedding are dominated by where the camera looks. Measured head orientation explains them on Ego-Exo4D for egocentric (R^2 0.49–0.70) but not exocentric cameras, and the scene layout explains them on WEAR. The part that the limbs' inertial signals predict separates activities across people. In a set-level probe, the rest separates exercise sets within a participant but not activities across people. This is consistent with separate roles: the video finds the clips of the same set, and the inertial classifier labels them. On WEAR, with temporal order removed and one random limb per one-second clip, we propagate inertial predictions over a within-participant video graph whose leading components are removed. This raises subject-wise cross-validated macro-F1 from 0.648 to 0.852. In

our comparisons, propagation helps only over the video graph, and fusing the modalities in one classifier, or adding inertial edges to the graph, performs worse.

Additional Key Words and Phrases: egocentric video, inertial sensing, human activity recognition, transductive inference, label propagation

1 Introduction

Video from a head-worn camera is now routinely turned into features by pretrained encoders. Generic encoders trained on web images and third-person video are applied to it directly. Large egocentric corpora such as Ego4D [18] have also produced encoders pretrained on first-person footage, such as EgoVLP [19], LaViLa [20] and, for embodied agents, VC-1 [23]. These encoders are built for egocentric tasks and judged by their downstream accuracy. Little attention has been paid to what their embeddings represent when the camera sits on the head, in particular how much of them reflects where the head points rather than what the wearer does. The question matters once such features are combined with body-worn inertial sensors. These are increasingly recorded together with egocentric video, in datasets such as WEAR [1] and Ego-Exo4D [10], and a head camera rarely shows the wearer's body, only what the head points at. We analyse what egocentric video embeddings encode and, from that analysis, propose how to use them together with inertial sensing for human activity recognition (HAR).

Inertial-based HAR assigns an activity label to each sliding window of an accelerometer stream. An important use case is the offline annotation of recorded wearable data. Annotation is a documented bottleneck of the field. Labelling the Opportunity footage took 14–20 times its duration [45], and the trade-off between label completeness and accuracy is studied in its own right [46]. Vision foundation models applied to the accompanying egocentric video have been used to reduce this effort [2]. In this offline setting the recording is complete before inference begins, so all clips of a participant can be processed together. Video is often available at collection time even when the deployed sensor is inertial, and the placement of the device is not always under the experimenter's control.

Existing HAR methods, including those benchmarked on WEAR [1], classify clips in their temporal context. A recent protocol on WEAR [4] removes that context. Its test set is a shuffled bag of one-second clips from four unseen participants. Each clip carries the accelerometer of exactly *one* randomly assigned limb (Figure 2) together with egocentric video features, and the stated aim is "to explore how to most effectively combine egocentric cameras with inertial sensors". Two obstacles follow. First, the per-clip inertial evidence is thin, because a single limb over one second often does not move during an exercise that moves another limb. Second, the temporal context that would normally resolve such clips is gone, and the order cannot be recovered from the video (Section 4).

We start from the observation that the protocol withholds order but not neighbourhood. Activity sets last tens of seconds, so each contributes many test clips spread across all four limb placements. Pooling the per-clip predictions within the true sets would raise macro-F1 from 0.648 to 0.933 without changing the classifier. Set membership is unavailable, but it can be read off from how the data arise. A set fixes both the activity and the circumstances in which it is performed, namely the place, the posture, the direction the participant faces and the scene in view. The activity drives the motion of the limbs and, through the body, part of what the head camera sees. The circumstances drive most of the rest of the view and are shared by every clip of the set, whichever limb carries its sensor. We use this relation in reverse (Figure 1b). The video identifies clips recorded under the same circumstances, that is, clips of the same set, and the inertial classifier labels them. Propagating the inertial predictions over a video graph then pools the four limbs over which a set's clips are spread. We remove the leading directions of the video embedding, which mix the view with the

appearance of the activity, before building the graph. We use no test labels and no temporal order (Section 5).

Our contributions are three-fold:

(1) *What egocentric video embeddings encode, and what degrades a graph built on them.* The leading principal directions of pretrained encoders applied to head-camera footage are dominated by where the camera looks. On Ego-Exo4D [10] they follow head orientation for four encoders, including the egocentric VC-1, but not for exocentric cameras. On WEAR they follow the scene layout. Removing them improves a within-participant graph for every encoder. Our analyses are consistent with an account in which the graph benefits from the view, which identifies the set, whereas the limb-predictable part in the same subspace degrades it (Sections 3 and 7.2).

(2) *Evidence that, in the configurations we compared, building the graph from the video and starting propagation from inertial predictions works better than the fusion variants we tried.* In a controlled comparison with the same inertial classifier, the graph source sets the sign of the effect (−0.060 on an inertial graph, +0.167 on the video graph). A published test-time procedure [30] changes macro-F1 by +0.001 on the classifier's own features and by +0.118 on the video graph. Every fusion variant we tried performs worse, although none was tuned as extensively as the proposed system (Sections 7.3 and 7.4).

(3) *A simple instantiation and what it shows about the evaluation protocol.* Built from standard components, the method raises subject-wise cross-validated macro-F1 from 0.648 to 0.852 without temporal order or test labels. We also report two properties of the protocol rather than of the method (Sections 5, 7.1 and 8).

The remainder of the paper is organised as follows. Section 2 reviews related work, Section 3 analyses what egocentric video embeddings encode, and Section 4 defines the recognition setting. Section 5 describes the approach built on that analysis and Section 6 the experimental setup. Section 7 reports results, and Section 8 discusses their meaning and implications for the evaluation protocol. Sections 9 and 10 state limitations and conclude. Additional analyses, including the complete list of tested explanations, are in the appendix.

2 Related Work

2.1 Representations of egocentric video

Egocentric video was first encoded with models pretrained on web images and third-person video, and the strongest video–language models relied on third-person corpora. Large egocentric corpora such as Ego4D [18] made pretraining on first-person footage possible. EgoVLP [19] adapts video–language contrastive learning to egocentric clips, LaViLa [20] trains on narrations that a language model generates for them, and VC-1 [23] learns visual representations for embodied agents by masked auto-encoding on egocentric video. These encoders target egocentric tasks such as video–text retrieval, action recognition, episodic memory and robot control, and are evaluated by downstream accuracy. Generic encoders such as VideoMAEv2, CLIP and DINOv2 remain in common use. Head motion, a defining trait of such footage, has been exploited as a cue, for example by EgoDistill [21], which reconstructs video features cheaply from head IMU signals. Probing studies ask what pretrained embeddings encode, for example properties of 3D scenes [22]. For word embeddings, the leading principal directions are known to dominate similarities, and removing them improves the representation [47], in relation to hubness [48]. We know of no work that relates the leading directions of egocentric video embeddings to the orientation of the head, or asks how such embeddings should be used when they are combined with body-worn sensors.

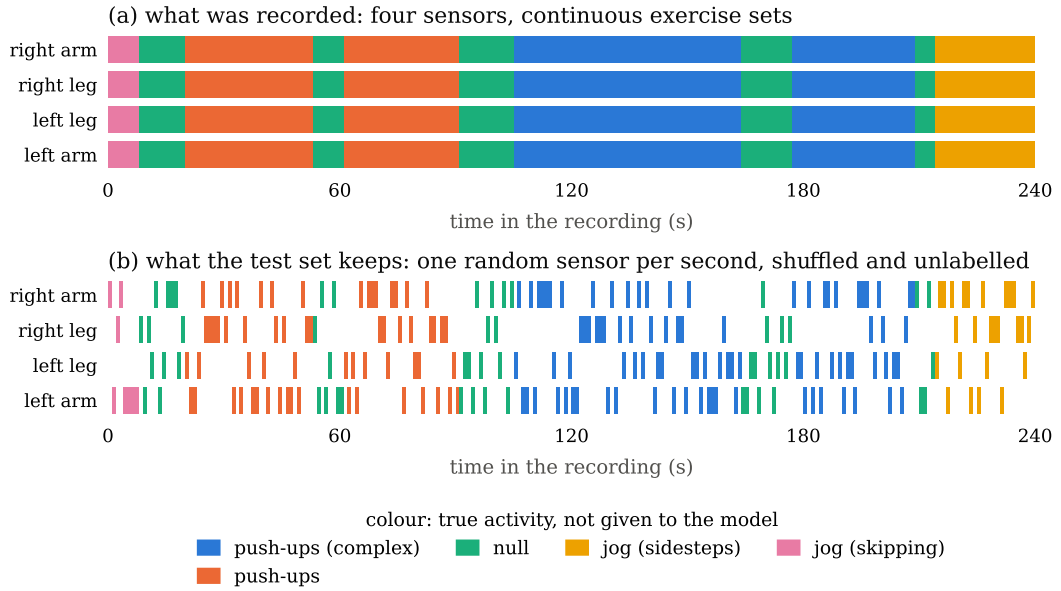


Fig. 2. The setting on real data (240 s of participant 0, coloured by the true activity). (a) What was recorded: all four limb sensors and activity sets lasting tens of seconds. (b) What the test set retains: each second retains *one* randomly assigned sensor, so the clips of one set come from different limbs. The clips are given shuffled and without labels. A set still leaves many clips with similar video, and the affinity graph exploits these.

2.2 Inertial HAR and the WEAR benchmark

WEAR [1] pairs four limb-worn accelerometers with a head-mounted camera. It benchmarks inertial, video and fusion models for activity recognition of its eighteen workout activities and for temporal action localisation, under leave-one-subject-out (LOSO) cross-validation. Temporal action localisation has since been shown to apply to inertial data on its own [3]. The two earlier editions of the challenge were inertial-only. Van Der Donckt et al. [11] used all four limb sensors with temporal context and reached 0.90–0.92 macro-F1, exploiting left–right symmetry and upper–lower limb pairing. The second edition restricted inference to a single sensor. Phan et al. [12] combined hand-crafted features with sensor-rotation augmentation and a gradient-boosting ensemble (0.588 macro-F1), and other teams addressed the dominance of the NULL class [13] and used reservoir computing [14]. Test splits differ between editions, so these numbers serve to locate results and do not allow a direct comparison. Inertial features are sensitive to where the device is worn. Existing remedies make the classifier insensitive through orientation-invariant features, symmetric limb pairing [11] or large multi-placement corpora [15]. Training with four sensors and testing with one is an instance of learning under privileged information [16].

2.3 Egocentric video and inertial sensing

Recent work couples egocentric video and inertial data to improve or compress the *classifier*. COMODO [17] distils a frozen video encoder into an IMU encoder so that inference can run on the IMU only. These are cross-modal transfers at training time in an inductive setting. Closest to our use of the video, Bock et al. [2] embed the egocentric video of WEAR and two other datasets with vision foundation models such as CLIP and cluster the embeddings. An annotator labels the centroid clip of each cluster, and this label is propagated to the corresponding inertial data. We share

the premise that egocentric embeddings group clips of the same activity. However, we propagate the predictions of an inertial classifier instead of human labels, at test time within each unseen participant, and we ask which part of the embedding should be removed before grouping. We also assign the modalities different roles at inference time. In our setting a privileged four-sensor teacher brings no improvement once the transductive structure is in place (Section 7.3). The video features we use are frozen VideoMAEv2 [24] embeddings distributed with the dataset.

2.4 Transduction and test-time adaptation

Adapting to an unseen person with unlabelled test data is an established setting for wearable HAR. Wang et al. [25] adapt without back-propagation, and SIGHT [26] exploits the temporal structure between consecutive clips, which our protocol removes. Label propagation on a k-NN graph is classical [27] and remains competitive with deep features [28]. Transductive few-shot methods such as TPN [29] learn the propagation itself, and LAME [30] adjusts a black-box classifier's outputs at test time with a k-NN Laplacian and no labels. In these methods the affinity graph is built from the same representation as the classifier, and in label propagation the graph contains labelled nodes. Our graph has no labelled nodes. It smooths predictions among unlabelled test clips and is built from a different modality. Mutual k-NN graphs follow the k-reciprocal idea of re-ranking in retrieval [31].

2.5 Smoothing predictions over another source's neighbourhoods

Smoothing a classifier's per-sample posteriors over a neighbourhood structure derived from a different source has a long history outside HAR. In hyperspectral remote sensing, pixelwise SVM probabilities from the spectrum are regularised by a Markov random field over spatial position [32] or by edge-preserving filtering guided by the image's leading principal components [33]. The dense CRF [34] and the bilateral solver [35] smooth network outputs with affinities from a reference image. In graph learning, Correct-and-Smooth [36] and APPNP [37] propagate the predictions of a structure-agnostic base model over a given graph, and population-graph GCNs [38] assign edges and node features to different sources by design. When several views are available, graph-based learning usually *fuses* them. Examples are sparse integration of view Laplacians [39], auto-weighting by the smoothness of the current solution under each view [40] and per-sample modality weights as in the weighted nearest neighbours of multimodal single-cell analysis [41]. Our contribution is a controlled comparison in a wearable setting. It shows that fusion of either kind performs worse than a division of roles in which the video builds the graph and inertial information enters it only through predictions.

3 What Egocentric Video Embeddings Encode

In this section we ask what a pretrained encoder represents when it is applied to head-camera footage. The egocentric pretraining work of Section 2 leaves this question open. The question does not depend on the recognition task, so we answer it on the recordings themselves, before we define the task in Section 4.

3.1 Data and encoders

Table 1 summarises the data. WEAR [1] has 22 training participants and 24 recordings of 18 outdoor workout activities. It provides 3-axis accelerometers at 50 Hz on both wrists and both ankles and 768-dimensional VideoMAEv2 frame embeddings at 30 fps from a head-mounted camera. Each exercise is performed in sets, which are continuous bouts at one place with a median length of 32 s. Time not annotated as an activity, such as breaks between sets or walking to the next station, is labelled NULL. We cut each recording into one-second clips. A clip holds the video features and the

Table 1. Datasets. WEAR is used for the method and every evaluation. Ego-Exo4D is used only for the analysis of Section 3.

	WEAR, training	WEAR, test	Ego-Exo4D (analysis only)
Participants	22 (24 recordings)	4, unseen	6 (79 takes)
Activities	18 outdoor workouts and NULL	same	basketball, dance, soccer
Inertial	3-axis acc. at 50 Hz on both wrists and ankles	one randomly assigned limb per clip	head trajectory (6 DoF) of the Aria glasses
Video	head camera: VideoMAEv2 embeddings; public raw video	VideoMAEv2 embeddings	egocentric (Aria) and one exocentric camera
Clips	about 69,000 one-second tiles, in order	12,234, shuffled	8,163 one-second clips
Labels	per second	hidden	not used

accelerometer signal of that second, and it is the unit that is classified. In this section we analyse the clips of the training participants.

Besides the VideoMAEv2 embeddings of the dataset, we use the CLIP ViT-B/32 [5] and DINOv2 ViT-B/14 [6] image encoders and VC-1 ViT-B/16 [23], which is pretrained by masked auto-encoding on over 4,000 hours of egocentric video and ImageNet. We apply these encoders to the public WEAR videos of the training participants (Appendix A). WEAR has no head sensor. To relate the embedding to the head, we use Ego-Exo4D [10], whose head-mounted Aria glasses record a six-degree-of-freedom head trajectory. We take two participants from each of three whole-body domains (basketball, dance, soccer; 79 takes, 8,163 one-second clips). We encode their egocentric video with the same four encoders and the extraction specification of the WEAR features. Per clip, we compute the head tilt (gravity direction in the device frame), head height, and rotation and translation speed. For comparison we encode the first exocentric camera of the same takes in the same way. We use Ego-Exo4D for this analysis only, and no part of the method is trained or tuned on it.

3.2 The leading components track head orientation on Ego-Exo4D

A head camera never shows the performer's body as a whole. The frame contains what the head points at, such as the ground, trees and sky, buildings, and the wearer's hands and feet. On Ego-Exo4D the head trajectory is measured. There, head orientation (tilt, height and heading) predicts the first two components of the egocentric embedding within a take with R^2 0.49–0.70 across the four encoders. The egocentric VC-1 gives the highest value (Table 2). Head tilt is linearly decodable from the top-30 subspace, better than from 30 random directions (0.74–0.91 against 0.61–0.79). Tilt is thus readable throughout the embedding but most strongly from the leading subspace. None of this holds for the exocentric cameras of the same takes encoded in the same way. For them, head orientation predicts the first two components with $R^2 \leq 0.20$, and tilt is only slightly more decodable from the top-30 subspace than from 30 random directions. The exocentric camera sees the same movements from a fixed position, so this control separates what the viewpoint contributes from what the activity itself contributes. All four encoders concentrate the view in the leading directions on egocentric footage, and none does so on the exocentric footage of the same takes. This concentration therefore appears to be a property of the head-mounted viewpoint rather than of a particular encoder.

Figure 3 shows this for one participant. When coloured by measured head tilt, the egocentric embedding is ordered from looking down to looking up along its first component. The exocentric

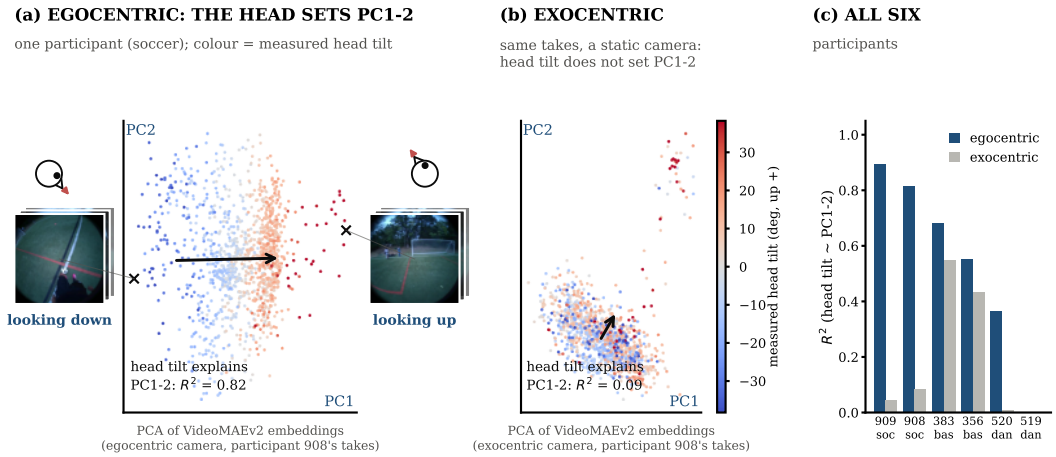


Fig. 3. What the leading components of an egocentric embedding follow, on Ego-Exo4D (VideoMAEv2, centred within each take). (a) Egocentric camera of one participant (soccer), coloured by the measured head tilt. The components follow the tilt. The arrow is the direction in which the tilt increases, with length proportional to the correlation, and the frames are the clips at its two ends. (b) Exocentric camera of the same takes. Here the components do not follow the tilt. (c) R^2 of head tilt from components 1–2 for all six participants. Frames from Ego-Exo4D [10], used under the Ego-Exo4D licence agreement, downscaled.

embedding of the same takes shows no such order. Within takes, head tilt explains components 1–2 of the egocentric camera with R^2 between 0.00 and 0.90 across the six participants (median 0.62; near zero for one dance participant). These values are in-sample, unlike the cross-validated values of Table 2. For the exocentric camera the value is near zero except in basketball (0.43 and 0.55), where the head moves with a visible body motion such as a jump shot. The direction that encodes head tilt is shared across people. A linear read-out trained on five participants predicts the sixth participant's tilt with correlation 0.87–0.97 for VideoMAEv2 (0.79–0.98 across the four encoders).

3.3 Scene layout and head tilt on WEAR

WEAR has no head sensor, so we measure what the view contains. We take the spatial layout of scene classes from a semantic segmentation of each clip's frame (Appendix A). Within a recording, this layout predicts components 1–2 with R^2 0.52–0.61, about as well as for the egocentric Ego-Exo4D video (0.60–0.81). The wearer's visible body parts reach 0.05–0.08, and nothing predicts components beyond the tenth. The layout itself is not specific to the egocentric view, because a person moving in a static exocentric frame also changes it (0.44–0.54 for CLIP, DINOv2 and VC-1. For VideoMAEv2 the cross-validated R^2 is −0.23, below that of predicting the mean). The specific part is that in the egocentric view the layout changes because the head moves. We train the tilt read-out on all of Ego-Exo4D and apply it to WEAR, which was recorded with a different head camera. Its output agrees with the per-frame camera pitch estimated there by GeoCalib [7] (median correlation per recording 0.58, 0.70 and 0.79 for VideoMAEv2, CLIP and DINOv2). It also ranks lying exercises as the most upward-looking. However, tilt accounts for only part of components 1–2 on WEAR (median R^2 0.10–0.34). There, the leading components depend as much on where the wearer is and which way they face.

Table 2. The leading components of egocentric and exocentric embeddings. Layout and Orient.: within-recording R^2 of predicting components 1–2 from the scene layout of the frame or from head tilt, height and heading (cross-validated over five time blocks). Tilt←top-30: R^2 of decoding head tilt from the top-30 subspace (cross-validated over takes), with 30 random directions in brackets. WEAR has no head sensor.

Encoder	Data, view	Layout	Orient.	Tilt←top-30
VideoMAEv2	WEAR, ego	0.52	—	—
	Ego-Exo4D, ego	0.60	0.49	0.74 (0.63)
	Ego-Exo4D, exo	−0.23	0.07	0.36 (0.27)
CLIP	WEAR, ego	0.61	—	—
	Ego-Exo4D, ego	0.64	0.51	0.78 (0.61)
	Ego-Exo4D, exo	0.44	0.05	0.22 (0.20)
DINOv2	WEAR, ego	0.58	—	—
	Ego-Exo4D, ego	0.72	0.58	0.84 (0.71)
	Ego-Exo4D, exo	0.46	0.20	0.33 (0.25)
VC-1	WEAR, ego	0.53	—	—
	Ego-Exo4D, ego	0.81	0.70	0.91 (0.79)
	Ego-Exo4D, exo	0.54	−0.14	0.20 (0.16)

3.4 The view marks the set rather than the activity

We test this division as follows. We split each embedding into its limb-predicted component, predicted out of fold from the four limbs' inertial features, and the rest. We then average both over each set (608 activity sets of at least ten clips). Below we refer to the two parts as the appearance of the activity and the circumstances. These names are interpretations. The split is defined by what the four limbs predict, it does not identify independent factors, and what four limbs predict need not be what a single-limb classifier already uses. Between sets of different participants, whose place, day and person differ, the limb-predicted component still separates the activities. The probability that a pair of different activities lies farther apart than a pair of the same activity is 0.91 (95% interval over participants $[0.88, 0.93]$). For the rest, the probe detects no activity separation (0.52, $[0.47, 0.58]$). It compares set averages by cosine distance, so it does not exclude activity information that a non-linear read-out or another metric could extract. Within a participant the rest separates sets instead. Two halves of one set lie 0.08 apart in cosine distance, as do adjacent sets of the same exercise. Adjacent sets of different exercises lie 0.93 apart, nearly as far as sets of another participant (0.99), whatever the gap between them. A new exercise comes with a new posture and facing, so the circumstances belong to the set. In this probe they do not indicate which activity it is. CLIP, DINOv2 and VC-1 give the same pattern. By construction the rest contains what the limbs do not predict, so we expect it to show little activity separation across people. The informative result is that it still separates sets within a participant.

3.5 Implications for combining the modalities

The leading directions of an egocentric embedding thus hold the view of the moment (where the head points, in which place and posture). Within that view they hold the appearance of the activity. The clips of a set share the view, and it changes from set to set. Linking clips with a similar view therefore finds the clips of the same set. In our probe the view shows no activity separation that transfers across people, so we do not rely on it to identify what they are doing. The appearance does encode the activity, but it is predictable from the limbs' inertial signals, and propagation pools

the inertial predictions of all four limbs within a set (Section 7.3). Seen from the head, an exercise also shares its appearance with the pauses around the set. These observations suggest a division of roles. The video builds the graph, the inertial classifier supplies the labels, and we remove the appearance from the graph. Section 5 builds a method on this division. Section 7.2 tests which part of the embedding the graph is better without, and Section 7.3 tests the division against fusion.

4 Problem Setting

Test protocol. We evaluate the method under a protocol on WEAR [4] that removes temporal order (Figure 2). The test set contains 12,234 one-second clips (windows in the challenge data) from four unseen participants. In the training recordings, NULL covers 40% of the clips (from 14% to 57% per recording). Each clip provides a $(50, 3)$ accelerometer segment from one limb with known placement. It also provides 15 rows of VideoMAEv2 embeddings, one per frame, each computed from the 16 frames around that frame (Appendix A). The clips are consecutive, non-overlapping one-second tiles of each recording, shuffled. Each second of a recording appears once with one randomly assigned limb.

Task and metric. The task is 19-way classification (18 activities and NULL) of each clip. It is evaluated by the macro F1-score over all 19 classes, including the NULL class. Segment-based metrics need start and end times and are undefined for a shuffled bag of clips. We therefore report record-based precision, recall and F1-score.

What the protocol removes. Temporal order cannot be restored from the video. We tried nearest-neighbour matching of clip-mean video features, directional matching from the last frame of one clip to the first frame of the next, and a global linear assignment with unique successors. All of them plateau at successor top-1 accuracy ≤ 0.35, which is too low for temporal models. Two properties of the data survive. Activity sets have median length 32 s, so each set contributes many clips spread over all four placements. In addition, the top-10 video neighbours of a clip share its label 77–88% of the time. The raw accelerometer signal does reveal adjacency for the quarter of consecutive pairs that happen to share a limb. We do not use it, and we discuss it in Section 8.2.

Transductive setting. Recordings are complete before inference, so we treat each test participant's clips jointly. All video features and model outputs of that participant may be used at test time, but test labels may not. Every model is trained on training participants only.

5 Method

The method implements the division of roles suggested by Section 3. The video builds a graph that links clips showing the same scene, that is, clips of the same set. The inertial classifier supplies the labels, and we remove the appearance of the activity from the graph. Figure 4 gives an overview. Per-clip classifiers produce class probabilities (Section 5.1). For each test participant we build a graph from the video embeddings (Section 5.2). We propagate the inertial predictions over it in stages that alternate with re-estimation of the graph (Section 5.3). Video evidence and a decision rule complete the pipeline (Section 5.4).

5.1 Per-clip classifiers

The inertial classifier produces $P_i \in \Delta^{19}$ for clip i from its single-limb accelerometer segment and placement. It is a weighted blend of two parts. The first is LightGBM on 78 hand-crafted features (per-axis moments, quantiles, zero-crossing rate, jerk statistics, orientation-invariant magnitude features, inter-axis correlations, spectral band powers and entropy) with a placement one-hot, at weight 1.5. The second is six 1D-CNNs on the raw segment with a placement embedding, at weight

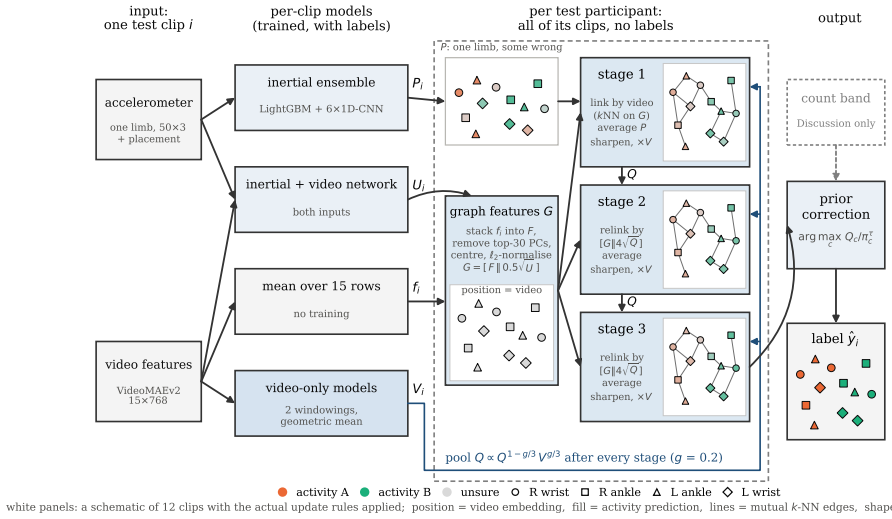


Fig. 4. Overview of the approach, from one test clip to its label. Left: the two inputs of a clip. Centre-left: per-clip models trained on the training participants. P, U and V are 19-class probabilities and f is the mean of the clip's 15 video rows. Centre-right: everything inside the dashed box runs once per test participant on all of that participant's clips and uses no labels. This includes the graph features G, three propagation stages with the graph rebuilt from $[G \,\|\, 4\sqrt{Q}]$ between stages, and the video-only probabilities V pooled after every stage. Right: the decision rule yields the label. The white panels are a schematic of a dozen clips with the actual update rules applied (position = video embedding, fill = activity prediction, grey = unsure, lines = mutual k-NN edges, shape = limb carrying the accelerometer). The inertial prediction P has wrong and unsure clips. Stage 1 links clips that look alike in video and averages their predictions, and stages 2–3 relink using the refined predictions as well. Raw inertial features never enter the graph. Inertial information enters it only through class probabilities (U in G and the refined Q), and the video never enters the inertial classifier. The dashed count band is the protocol constraint of Section 8.2 and is not part of the reported system.

1.0 each. Three of them use rotation augmentation (±20° to ±30°), and three additionally take the 78 features as an auxiliary branch. Each training clip is expanded into four samples, one per placement. Two further classifiers use the video. A network on both modalities gives probabilities U_i that enter the graph features. A video-only network gives probabilities V_i that are pooled after propagation. Architectural details are in Appendix A.

5.2 Participant-level video graph

For each test participant, let $f_i \in \mathbb{R}^{768}$ be the mean of clip i's 15 video rows and F the matrix of these vectors. We remove the top $m = 30$ principal components of F, computed within the participant, then centre and ℓ_2-normalise the rows. The graph features are

$$G_i = \left[\, F_i \,\middle\|\, \gamma\sqrt{U_i} \,\right], \qquad \gamma = 0.5, \tag{1}$$

re-centred and row-normalised. We connect each clip to its $k = 30$ nearest neighbours by cosine similarity s_{ij} and retain only mutual edges. A k-NN graph needs neither the number of sets nor their boundaries, which are unknown, and k is close to the median set length of 32 clips. Hard clustering did no better (Appendix F). The affinity is $W_{ij} \propto \exp(10\, s_{ij})$ over retained edges, row-normalised. All operations use only the participant's own clips, with no temporal order and no labels. Section 3 motivates the component removal and Section 7.2 analyses it.

5.3 Propagation with graph re-estimation

Let $P \in \mathbb{R}^{n\times 19}$ collect the participant's inertial predictions. Each stage propagates

$$Q^{(t+1)} = (1-\alpha)\,Q^{(0)} + \alpha\,WQ^{(t)}, \qquad \alpha = 0.75, \tag{2}$$

for five iterations from $Q^{(0)}$, which is P in the first stage and the output of the previous stage afterwards. It then sharpens the result, $Q \leftarrow Q^{1.4}$ renormalised. Between stages the graph is rebuilt from $[\, G \,\|\, \lambda\sqrt{Q}\,]$ with $\lambda = 4$, so that clips whose refined predictions disagree are pushed apart. We use three stages. The stages therefore alternate between estimating the labels given the graph and re-estimating the graph given the labels. Section 7.4 tests this reading.

5.4 Video evidence and decision rule

After every stage the video-only probabilities are pooled log-linearly,

$$Q \;\propto\; Q^{1-g/3}\,V^{g/3}, \qquad g = 0.2, \tag{3}$$

so that the total video weight is g over the three stages. Macro-F1 favours recall on rare classes, so the posterior mode is not the best decision. The final label is therefore $\hat{y}_i = \arg\max_c Q_{ic}/\pi_c^{\tau}$, where π_c is the mean predicted probability of class c over all clips being labelled (at test time all test clips, in cross-validation the held-out clips of all participants) and τ is a single scalar selected by cross-validation (0.9 for the final system). Averaging π_c over each participant's clips instead gives higher cross-validated scores (Section 7.5), but we report the rule that we fixed before the hidden test and used there.

6 Experimental Setup

Evaluation protocol. We use five-fold cross-validation grouped by participant, with the two recordings of one person in the same fold. The benchmark's leave-one-subject-out protocol is its $k = n$ limit. We build the graph and propagate per held-out participant, as at test time. The class prior of the decision rule is averaged over all clips being labelled (Section 5.4). The validation data mirror the test structure, with non-overlapping one-second tiles that each have one uniformly random limb. Because that assignment is random, we repeat it over seeds (three for the main results, two for the comparisons unless stated) and report means. The seed-to-seed standard deviation is about 0.002, and we treat differences below 0.005 as inconclusive. For the main gains we also report 95% intervals from resampling the 22 participants (2,000 draws), which capture the variation between people. We selected all hyperparameters on this cross-validation. The training epoch of each network and the threshold τ of each configuration are also chosen on the held-out folds, with the labels against which we report. The reported scores are therefore the cross-validated results of an exploratory design process, not unbiased estimates of generalisation, and the intervals do not include the uncertainty of model selection. The seeds repeat the limb assignment, not the selection. The hidden test set of the challenge (Section 7.1) is the only evaluation outside this loop.

Compared variants. Beyond component ablations, we compare against the following. (i) Fusion at the classifier: a base that consumes both modalities, and a blend of the two bases. (ii) Fusion at the graph: adding inertial k-NN edges, learning view weights by the auto-weighting criterion of [40], and per-clip weights in the style of [41]. (iii) A different propagation rule, LAME [30], on either graph. (iv) A four-sensor inertial teacher, used directly or by distillation. (v) A graph pooled across participants. Models that need an ordered stream do not apply. Throughout, fusion means combining the two modalities in one classifier or in the edges of the graph. The method itself also uses both modalities: it adds the probabilities of video models to the graph features and pools them after every stage (Section 5). The comparison is therefore not between one modality and two, but

between where the video enters. In the method it builds the graph and enters the labels only as a weighted prediction, whereas in the variants it enters the classifier's input or the graph's edges. The comparison is not matched in tuning effort. The inertial base is an ensemble of LightGBM and six networks, the classifier on both modalities is a single network with its own training schedule, and the variants reuse the propagation settings tuned for the proposed system.

Use of AI tools. The code that implements the methods, experiments and analyses was written, and the figures were produced, with an AI coding assistant (Claude Code, Anthropic) under the author's direction. The author designed the study and the analyses, checked every reported number against the code to be released, and takes responsibility for all content.

Data and ethics. We use only public datasets (WEAR and Ego-Exo4D), which their authors collected with the participants' consent and which we use under their licences. No new data were collected. The figures show the scene and the wearer's own limbs. Test labels were never used, all hyperparameters were selected on subject-wise cross-validation, and the transductive steps use only test-set features and model outputs. Raw video was used for training participants only.

7 Results

We report the results in five parts. Part (1) gives the gain of each component of the method (Section 7.1). Parts (2) and (3) test the analysis of Section 3. They ask what component removal takes out of the video embedding (Section 7.2) and whether the modalities should be fused or given separate roles (Section 7.3). Part (4) covers what the propagation stages do (Section 7.4). Part (5) covers robustness to unseen participants, to clips that arrive in parts and to the hyperparameters (Section 7.5).

7.1 Main results

Table 3 reports record-based precision, recall and F1-score for the successive components, split into the NULL class and the 18 activities. The inertial classifier reaches 0.648 macro-F1 on single clips. The earlier single-sensor result on WEAR, 0.588 [12], was obtained on a different test split. Propagation over the video graph raises macro-F1 by +0.182, as do pooling the video-only probabilities (+0.013) and removing the leading components of the video features (+0.009), which gives 0.852. When we resample participants, the 95% intervals of the three gains are [+0.167, +0.196], [+0.009, +0.017] and [+0.005, +0.013]. The three gains hold for 22, 19 and 18 of the 22 participants. Recall increases more than precision, as the decision rule intends. The transductive stages benefit the activities far more than NULL, and the gap between F_1^{NULL} and F_1^{act} shrinks from 0.119 to 0.026. The confusion matrices (Figure 5) show where this happens. The inertial classifier's dominant error is activity → NULL, because a limb at rest during an exercise that moves another limb is inertially indistinguishable from a break. Propagation removes most of that column because other clips of the same set, recorded on other limbs, outvote it.

Pooling the inertial predictions within the true activity sets gives an upper bound of 0.933. The approach closes 72% of the gap between the inertial classifier and this bound. Per class (Appendix B), the largest gains are on classes that a single placement cannot capture. A leg sensor all but misses arm stretches (recall 0.05 for the shoulder stretch and 0.10 for the triceps stretch), yet the final system reaches F1 0.80 and 0.94 on them. This is because a set contains clips from all four limbs, and propagation transfers the evidence of the arm clips to the leg clips. The remaining errors concentrate in pairs of variants of one exercise, such as plain and complex push-ups. These look the same from the head and differ only in how the limbs move.

On the four hidden test participants, the configuration of Table 3 reaches 0.871 macro-F1 on the part of the test set that the challenge scores publicly [4].

Table 3. Record-based results under subject-wise cross-validation (mean over three seeds). F_1 is macro-F1 over all 19 classes. The last two columns split it into the NULL class and the 18 activities.

	P	R	F_1	F_1^{NULL}	F_1^{act}
Inertial classifier	0.658	0.644	0.648	0.761	0.642
+ propagation	0.825	0.840	0.830	0.861	0.828
+ video pooling	0.836	0.854	0.843	0.872	0.841
+ component removal	**0.842**	**0.865**	**0.852**	**0.876**	**0.850**

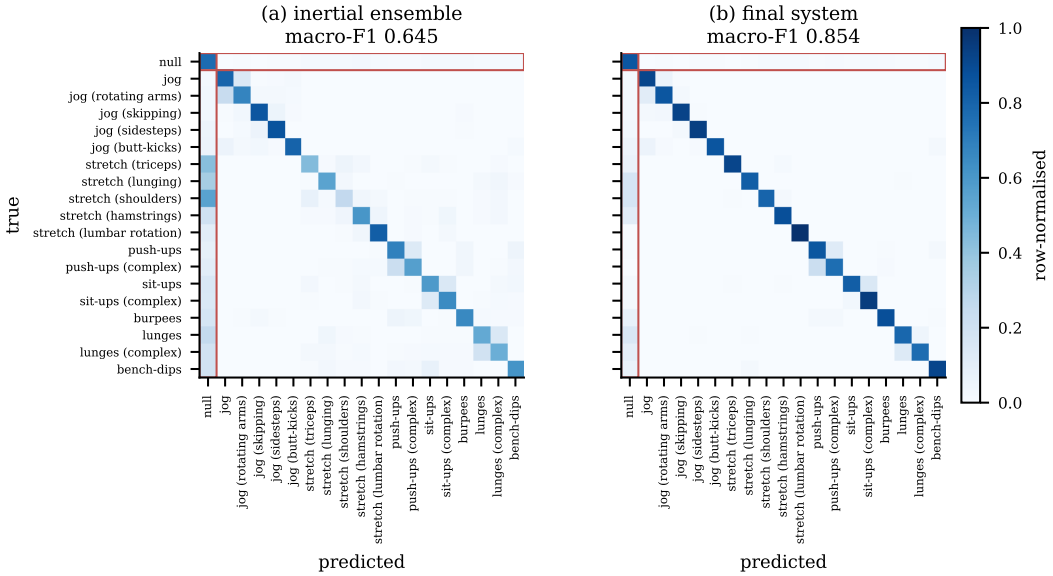


Fig. 5. Row-normalised confusion matrices before and after the transductive stages (seed 42). The NULL row and column are outlined. The inertial classifier's dominant error is activity → NULL, and propagation removes most of that column.

7.2 What component removal takes out

Removing the leading principal components raises macro-F1, although they are the most discriminative. Removing the top m principal components of the per-participant video embedding before building the graph raises macro-F1 (Figure 6b), with an interior optimum at $m = 30$ (0.8429 ± 0.0028 at $m = 0$, 0.8516 ± 0.0018 at 30, 0.8434 at 80 and 0.8325 at 120). The removed components are not a nuisance subspace. Within a participant, components 1–30 contain 77.2% of the variance and a mean activity η^2 of 0.266, against 0.047 for components 31–100 (Figure 6a). They also contain the activity information that transfers between participants. Retrieving a clip's neighbours from *other* participants gives label purity 0.408 on the raw features and 0.198 after removing 100 components (chance 0.161). Neighbourhood purity within a participant also falls (0.732 at $m = 0$, 0.633 at $m = 30$), so standard graph-quality measures predict the opposite of what the system does. The removal operates on directions rather than on coordinates. Deleting 400 of the 768 channels changes macro-F1 by only +0.0027. Truncated PCA whitening, which keeps the top 128 or 256 directions and equalises their variance, performs comparably to removal (0.8527 and 0.8535, within

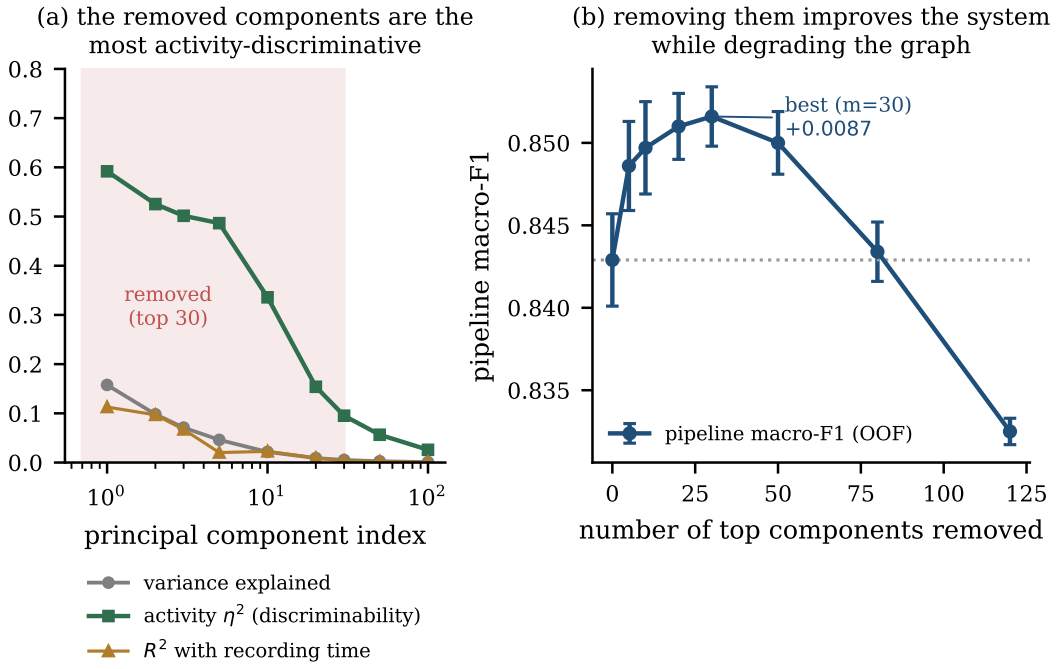


Fig. 6. (a) The leading components contain both the most variance and by far the most activity discriminability. (b) Removing them nevertheless raises macro-F1, with an interior optimum at 30, while neighbourhood activity purity falls (0.73 at $m = 0$, 0.63 at 30, 0.56 at 50).

our threshold of 0.005). The same truncation without whitening does not (0.8415 and 0.8425). This suggests that the weighting of directions in the similarities matters, but it does not separate removing specific content from correcting anisotropy. The analyses below describe what the leading directions contain (Appendix D, item 9).

The effect of removal is consistent across encoders. We encoded the public WEAR videos of the training participants with CLIP [5], DINOv2 [6] and VC-1 [23] (eight frames per clip, see Appendix A). We replaced only the features that the graph is built from and kept every classifier fixed. Removing the top 30 components increases macro-F1 for all four encoders in both seeds (Table 4). The gain is +0.007 to +0.015 with the video-probability block in the graph and +0.012 to +0.023 when the graph is built from the encoder's features only. For each encoder, $m = 30$ is the best setting tried. The improvement is largest for the encoder pretrained on egocentric video. The rest of this section asks what the removed directions contain and which part of that content the graph is better without.

The removed components are not limited to the view. If the gain came from discarding the view, subtracting the part of the embedding that the view explains should reproduce it. It does the opposite (Table 5). Subtracting the component explained by the scene layout, which holds half to two thirds of the variance, reduces macro-F1 by 0.006 to 0.018. Subtracting what the visible body explains changes nothing, and neither does subtracting what the camera tilt or the camera shake explains. We estimate the camera tilt from each frame with GeoCalib [7], which recovers the measured head tilt on Ego-Exo4D with $R^2 = 0.75$. GeoCalib is least reliable on views without a

Table 4. Macro-F1 on WEAR when only the graph's features are replaced (20 participants, 61,404 clips, mean of 2 seeds). The classifiers are fixed, and the per-clip prediction without a graph scores 0.644. "+prob.": the graph also contains the video-probability block. "features only": it does not.

	+prob.			features only		
Encoder	$m = 0$	$m = 30$	Δ	$m = 0$	$m = 30$	Δ
VideoMAEv2	0.8410	0.8480	+0.0070	0.8339	0.8455	+0.0116
CLIP	0.8342	0.8435	+0.0093	0.8175	0.8351	+0.0176
DINOv2	0.8337	0.8429	+0.0092	0.8165	0.8332	+0.0167
VC-1	0.8317	0.8462	+0.0145	0.8151	0.8384	+0.0233

Table 5. Change in macro-F1 on WEAR when a component is subtracted from the video features before the graph is built (features-only graph, 2 seeds). Components are the part of the embedding explained, within a participant, by the named quantity. *Uses ground-truth labels and is diagnostic only.

Subtracted	VideoMAEv2	CLIP	DINOv2	VC-1
top 30 components	+0.012	+0.018	+0.017	+0.023
components 1–2 only	+0.002	+0.004	+0.003	+0.004
scene layout	−0.013	−0.009	−0.006	−0.018
visible body	−0.002	−0.001	−0.001	−0.002
camera tilt	+0.000	+0.001	−0.000	−0.001
camera shake	+0.001	+0.000	+0.001	+0.000
what the four limbs predict	+0.010	+0.011	+0.010	+0.010
activity means*	+0.013	+0.022	+0.030	+0.017
what the classifier P predicts	−0.018	−0.019	−0.014	−0.025

horizon or straight edges, such as close views of the ground or the sky, and such views are frequent at the ends of the leading components (Figure 11). We measure camera shake by phase correlation, which recovers the measured inter-frame head rotation on Ego-Exo4D with $R^2 = 0.89$. Removing only components 1–2 yields a sixth to a quarter of the improvement from removing 1–30. The view lets the graph group clips recorded at the same place and moment, and the graph needs it. The same view also encodes a take's task across recordings on Ego-Exo4D. There, removal makes the transfer of labels between takes less accurate (0.744 → 0.724, 0.743 → 0.661, 0.740 → 0.713 at $m = 30$).

The removed components contain a part of the embedding that the limbs predict. The gain is reproduced by subtracting from each clip's embedding the out-of-fold prediction of that embedding from the four limbs' inertial features. With the features-only graph, this recovers 41–83% of the gain of removal (least for VC-1, whose embedding the limbs predict least). A control that subtracts the same predictions permuted among clips is worse than no subtraction. The limbs cannot predict the view. None of scene layout, visible body, camera tilt or camera shake explains the limb-predicted component ($R^2 \leq 0.05$). The limbs do, however, predict the activity. The per-participant activity means of the embedding hold a third of its variance, and 84–86% of that part is contained in the limb-predicted component (VideoMAEv2, CLIP, DINOv2). Subtracting the activity means themselves is a diagnostic that uses the labels. For the three generic encoders it matches or exceeds removing the

Table 6. Change in macro-F1 due to propagation against the same classifier without it (three seeds, configuration before component removal). Modality agreement is annotated. It does not predict the sign, but the graph source does.

	inertial graph	video graph
inertial classifier	−0.060 *(same)*	**+0.167** *(cross)*
video classifier	−0.016 *(cross)*	+0.037 *(same)*

top 30 components (+0.013, +0.022, +0.030 against +0.012, +0.018, +0.017), and for VC-1 it yields three quarters of that improvement (+0.017 against +0.023). The harm is not a correlation between the classifier's errors and the video. We shuffled the classifier's predictions among the clips of the same participant and class, which preserves its confusions and removes any link to the video. This leaves the gain in place and even enlarges it. On the other hand, subtracting what the classifier's own predictions explain reduces macro-F1, possibly because it imports the classifier's errors into the video.

Which clips removal corrects. Appendix D shows which clips removal corrects. These are mostly NULL clips captured by the activity next to them and static stretches captured by NULL. The appendix also compares removal with subtracting the limb-predicted component, which a test clip carrying one limb cannot compute. These analyses describe what removal changes. They do not establish that removing this content is what causes the gain, since whitening, which keeps the leading directions, gives a similar gain.

7.3 Role separation versus fusion

The graph source sets the sign. Table 6 crosses the classifier modality with the graph modality, measured at the configuration before component removal. It reports the change due to propagation against the same classifier without it. The video graph improves both classifiers, and the inertial graph degrades both. Whether the two modalities agree does not predict the sign. Propagation improves the same-modality pair (video, video) and degrades the cross-modality pair (video classifier, inertial graph). Agreement affects only the magnitude (the video graph raises the inertial classifier by +0.167 and the video classifier by +0.037). This is consistent with the two representations being close to orthogonal. Regressing inertial features on the video embedding explains only 0.21–0.25 of their variance, and the mean canonical correlation between the two spaces is 0.101.

The graph source has a larger effect than the propagation rule. Table 7 repeats the comparison with LAME [30], which adjusts classifier outputs as $z_i \propto p_i \odot \exp(\beta \sum_j W_{ij} z_j)$. In its original setting, with the graph built from the classifier's own (inertial) features, LAME changes macro-F1 by +0.001. The same LAME on the video graph yields +0.118, and our propagation rule on the video graph by +0.158. Changing the graph source changes macro-F1 by an order of magnitude more than changing between these two propagation rules.

Fusion performs worse in every form we tried. These variants were tuned less than the proposed system (Section 6), so the comparison shows which designs worked here rather than that fusion is worse in general. In the full system, a classifier consuming both modalities is weaker than the inertial one on its own (0.608 vs. 0.648) and far weaker after propagation (0.663 vs. 0.852). Blending the two gives the strongest single classifier (0.739), yet after propagation it too performs worse (0.783). One possible explanation is that fusing at the classifier puts the video's view of the activity into every clip's prediction, where the graph can no longer correct it. In a two-stage version of

Table 7. Graph source and fusion (single stage unless stated, two seeds). Base: the inertial classifier alone, 0.6485. Gains here and in Table 6 are measured in reduced configurations and are not directly comparable with those of Table 3.

Graph	Rule / fusion	F_1
inertial (classifier's features)	LAME, best β	0.6493
inertial (classifier's features)	ours	0.6025
video	LAME, best β	0.7668
video	ours	**0.8065**
full system, first-stage graph varied		
video only	—	**0.8523**
per-clip weights (WNN-style)	fusion	0.7856
uniform half	fusion	0.7627
inertial only	—	0.6836

the system without video pooling, adding inertial k-NN edges to the video graph degrades it monotonically (0.810 with the video graph only, 0.709 with 20 inertial edges per clip, one seed). Learning view weights by minimising the smoothness of the propagated distribution under each view's Laplacian [40] converges to an almost even split (0.518 video, 0.482 inertial). In a single-stage version of the system, these weights reduce macro-F1 by 0.067 relative to the video graph only. The criterion itself differs by only 11% between the views, while macro-F1 differs by 0.19. We also tried per-clip weights in the style of weighted nearest neighbours [41], in which each clip weights the two similarities by how well each modality's neighbours explain its own profile. These weights become sharply bimodal (mean video weight 0.61) and reduce macro-F1 by 0.067 (Table 7).

Propagation already performs the sensor fusion. A teacher that uses all four sensors is a much stronger clip classifier than the single-sensor inertial classifier (0.762 vs. 0.648 out-of-fold). Yet distilling it into one of the inertial networks changes the system by 0.0001, and using it directly as the base is worse than the weaker student (0.812 vs. 0.852). The reason is that propagation already integrates the four placements. Only 0.253 of a clip's video neighbours wear its limb, which is the chance rate, and restricting the graph to same-limb clips reduces macro-F1 by 0.118. Further analysis of the base classifier's contribution is in Appendix C.

7.4 What the stages do

Table 8 tests the reading of Section 5.3 by holding the sharpening and pooling schedule fixed and changing only what the graph is rebuilt from. More propagation on the same graph does not reproduce the gain of the stages. Repeating three rounds without rebuilding (-0.032) or running fifteen iterations in one round (-0.022) is no better than a single stage. Rebuilding improves the result, and the source of the rebuild determines the sign. With the refined Q, the fraction of edges joining clips of different true activities halves, from 0.194 to 0.087. With the unrefined inertial prediction P, it rises to 0.245 and macro-F1 decreases by 0.114. With a permuted Q, it collapses. Rebuilt from the true labels, the graph has no such edges and the system reaches 0.929, close to the level of the set oracle. The stages are therefore an alternating estimation of labels and affinity that works once the labels have been refined.

Table 8. What the stages do: macro-F1 (two seeds) and the fraction of edges joining clips of different true activities at stages 1/2/3. The sharpening and pooling schedule is identical in every row.

Graph between stages	F_1	Cross-activity edges
rebuilt from refined Q (ours)	**0.8523**	0.194 / 0.087 / 0.087
single stage	0.8303	0.194
not rebuilt, three rounds	0.8204	0.194 / 0.194 / 0.194
not rebuilt, 15 iterations	0.8299	0.194
rebuilt from base P	0.7379	0.194 / 0.245 / 0.245
rebuilt from permuted Q	0.7396	0.194 / 0.703 / 0.604
rebuilt from true labels (oracle)	0.9289	0.194 / 0.000 / 0.000

7.5 Robustness and sensitivity

Participant identity. The graph compares clips of one participant only. Participant-specific cues such as place, clothing and light therefore serve to recognise a set rather than act as a nuisance. A graph pooled across the four participants of a test-sized group, with no penalty on cross-participant pairs, forms only 5.1% cross-participant edges, of which 37% join different activities. It leaves the score unchanged (+0.0007). The per-clip classifiers are applied across participants. We measured their reliance on participant identity by training the same LightGBM on other sets of the *same* participant instead of on other participants. This raises macro-F1 by only +0.0014 on inertial features (0.6005 vs. 0.5991) and +0.009 on video features (0.4556 vs. 0.4465), with a per-participant range of −0.18 to +0.27 for video. The video, whose reliance varies more, enters only through the within-participant graph and the pooled probabilities.

Where unseen participants fail. Final macro-F1 ranges from 0.64 to 0.93 across the 22 participants. It tracks the inertial classifier's score ($r = 0.88$) and the set-oracle ceiling ($r = 0.87$) but not the purity of the video graph ($r = 0.12$). The lowest score is largely a labelling error. Public work [43] matched the training files to the per-session table of the WEAR supplementary material [1] and found that this participant's last session (1,419 s) is labelled NULL throughout although nine activities were recorded in it. The same holds for the last 112 s of another participant. Our system labels 62% of these 1,531 clips as activities, which the released labels count as errors. Without them, this participant scores 0.78, and the range becomes 0.72 to 0.93. Of the other three hardest participants, two fail at the NULL boundary, where gentle exercises and stretches are predicted as NULL, and the third in exercise variants. Deciding NULL separately has no benefit. The NULL class is detected worst from the video only (AUC 0.849), better from the inertial classifier (0.878) and best from the refined joint distribution (0.942). Every two-level rule we tried scores below the joint 19-way decision (−0.011 to −0.095).

Clips available in parts. The method treats a participant's clips jointly. If they become available in parts, as when a stream is processed every so often, the same system run separately on consecutive parts of the session retains most of its gain. It scores 0.853, 0.857 and 0.844 with halves, quarters and tenths of the session, against 0.852 with all clips and 0.648 for the inertial classifier. Each 95% interval of the difference from 0.852 includes zero. The parts are contiguous in time, so this models a deployment in which clips arrive in order. It does not model the shuffled test set. Only the graph and the propagation are restricted to the part. The class prior of the decision rule is still averaged over all clips, which such a deployment would not have.

Decision rule and video rows. Two definitions in our evaluation could have been made differently. Averaging the class prior π_c of the decision rule over each participant's clips, rather than over all clips being labelled, raises macro-F1 by +0.019 to +0.020 for every propagated configuration (three seeds; 0.8516 → 0.8707 for the final system) and by +0.002 for the inertial classifier. The gain from propagation therefore grows (+0.182 → +0.199), while the gains between the propagated configurations are nearly unchanged. We did not evaluate it on the hidden test set, where a per-participant class balancing that gained in cross-validation lost (Appendix F). Building the graph from the video rows that the test set supplies, rather than from the rows of Appendix A, changes the propagated configurations by +0.000 to +0.002 (0.8518 for the final system).

Sensitivity to hyperparameters. No one-at-a-time change of the hyperparameters improves macro-F1 on both seeds (Appendix A).

8 Discussion

8.1 What the graph means in the domain

Read in terms of the activity, the graph connects the moments of a participant's session in which the head camera saw the same thing: the same place, the same posture and the head pointing the same way. A set of one exercise lasts about half a minute (median 32 s) and holds all three roughly constant, so the graph is a stand-in for set membership. The graph is useful because a clip's video neighbours often carry sensors on other limbs. An ankle clip during an arm exercise contains almost no activity information, but its video neighbours include the same set seen from the other limbs (the neighbours of the clip in Figure 1 are spread 6/9/9/6 over the four placements). Propagation therefore performs at test time the four-sensor fusion that the protocol makes unavailable to each clip. When we rebuild the graph from refined predictions, "the same scene" narrows to "the same scene and apparently the same exercise". This separates exercises performed in similar postures. The video cannot supply the exercise variant. Plain and complex push-ups look the same from the head and differ in how the limbs move. Video thus supplies the neighbourhood structure, and inertial sensing supplies the class evidence.

Section 3 makes the same division inside the video embedding. Its leading directions hold two things. One is the view, which lets the graph group the moments of a session. The other is the appearance of the activity, which the limb sensors can predict. From the head, the appearance of an exercise is the appearance of its place and posture. The pauses around a set and the stillness of a static stretch share this appearance. The clips that removal corrects fit this picture: they are mostly pauses captured by the activity next to them and static stretches captured by NULL. Subtracting the limb-predicted part also reproduces much of the gain. Whitening the leading directions without discarding them does as well, however, so we cannot say whether the gain comes from removing this content or from changing how much the leading directions weigh in the similarities. The analysis of what the embedding contains stands on its own. That its removal causes the gain remains a hypothesis.

8.2 Implications for the evaluation protocol

Shuffling does not remove adjacency. The test clips are consecutive, non-overlapping tiles. When two consecutive clips carry the same limb (a quarter of pairs), the accelerometer signal continues across their boundary, and the successor of such a clip can often be found (Appendix G). The publicly released solution of K. Rudra builds on this signal [44]. Added on top of our system, it contributes little: smoothing the predictions along the recovered links improves cross-validation macro-F1 by only +0.004. Systems built around it gain much more. A public pipeline that recovers the order and propagates along the links scores 0.906 on the public leaderboard, and an extension

with learned links reaches 0.929 (nested cross-validation 0.930) [43], against 0.871 for our system (Section 7.1). The same work reports that replacing each link by a random clip of the same bout scores higher (0.958) than the true successor (0.929), which agrees with our view that what helps is finding clips of the same set, not the order. We do not use the links, because they are a property of how the test set was cut rather than of activity recognition. Our results should be read as what is achievable when order is genuinely absent. Leaving gaps between test clips in future editions would close this channel at no cost to the task.

Protocol knowledge as a test-time constraint. WEAR has a fixed protocol, in which each participant performs all eighteen activities for roughly 90–120 s. In the training recordings an activity class occupies a median of 94 one-second clips (5–95% range 78–126), and 431 of 432 recording–class pairs are non-empty. Yet 31% of the recording–class counts predicted by our system fall outside that range. Following the count calibration of the same solution [44], we add the protocol as a constraint. For each participant, we adjust per-class biases on the prior-corrected scores until every activity class has between *lo* and *hi* predicted clips. NULL absorbs the remainder, so its share stays free (it varies from 0.14 to 0.57 with recording length). With the range set from the *other* training recordings, this increases macro-F1 by $+0.0160 \pm 0.0012$ (three seeds, all positive). The largest gains are on lunges and push-ups. The constraint raises macro-F1 more for the four hardest participants than for the rest (+0.057 vs. +0.019), and it is most effective where variants are confused. Confusing two variants pushes one count above the range and the other below it. The constraint is close to class mass normalisation [42]. It differs from per-participant Sinkhorn balancing, which fixes every class including NULL and did not transfer (Appendix F). It uses no test labels but assumes that test participants followed the protocol. This holds for WEAR by design and not for free-living data.

9 Limitations

All results come from one dataset. The design choices were made on the same cross-validation that we report (Section 6), so the scores are optimistic by an amount we do not measure, and the hidden test set is the only independent check. All scores use the labels as released. Besides the two unlabelled stretches of Section 7.5, one session has its labels 10 s late [43]. Leaving out the former and moving the latter raises the final macro-F1 from 0.852 to 0.865 and the inertial classifier from 0.648 to 0.658 (three seeds), so the gains are unchanged. We keep the released labels, since the test labels presumably follow the same conventions. The video rows of our training clips are offset from the test rows by up to eight frames. Aligning the graph's rows changes nothing, but we did not retrain the classifiers on aligned rows (Appendix A). Cross-validation covers 22 participants and the hidden test four, and the variation between participants is large (Section 7.5). The approach makes two assumptions. First, a participant's clips are available together, although a tenth of a session at a time retains most of the gain (Section 7.5). Second, an activity set is visually consistent from the head. Free-living data, in which people move between places during an activity or do different activities in one place, may violate the second assumption. The 2×2 design of Table 6 confounds invariance with discriminativeness, since the video graph is also the stronger representation in absolute terms. The negative claim (modality agreement does not determine the sign) is therefore established more firmly than any positive one. Our account of component removal rests on measurements of the view made with off-the-shelf models (segmentation, single-image calibration, phase correlation). We validated these on Ego-Exo4D but not on WEAR. The component this account identifies cannot be computed at test time, where each clip carries one limb. We have not shown that removing it is what causes the gain: whitening the leading directions gives a similar gain, and the two explanations are not separated (Section 7.2). The Ego-Exo4D check covers four encoders, one of them pretrained on egocentric video. However, it uses our own encoding, a

different camera, sports rather than workouts, six participants and one exocentric camera per take. The encoder comparison on WEAR excludes three recordings (one that matched its public video only weakly and two whose public video we did not download). Whether removal improves results on egocentric tasks other than activity recognition is untested.

10 Conclusion

The leading directions of egocentric video embeddings from pretrained encoders primarily reflect where the camera looks. We observe this across four encoders, including one pretrained on egocentric video, and not for exocentric cameras. It reflects what a head camera records: where, and in which posture, an activity takes place, and far less of the movement that makes it that activity. For combining such video with body-worn sensors, this suggests a division of roles, and our comparisons are consistent with it. In this setting, the video is most useful for finding the clips of the same set, and the inertial classifier for labelling them. The view makes the video a good grouping signal. Reducing the dominance of the leading directions, by removing or whitening them, improves the graph further. Whether this works by removing the appearance of the activity that comes with the view, or by re-weighting the directions in the similarities, we have not separated. Every fusion variant we tried, in one classifier or in the graph edges, performed worse than the division of roles, although none was tuned as extensively as the proposed system. Re-estimating the graph from refined labels makes a multi-stage schedule effective. The propagation that implements the division is not new. What we add is evidence on when it works in this setting, and an analysis of what the removed components contain. The analysis also exposes two properties of the benchmark, recoverable adjacency and a strong class-count prior, that future shuffled-clip evaluations should take into account.

Code availability. The code that reproduces every table and figure is available under the MIT Licence at https://github.com/habakan/what-egocentric-embeddings-encode.

A Implementation Details

Classifiers. The 1D-CNNs use four convolutional blocks (kernel sizes 7, 5, 3, 3 and $64 \rightarrow 128$ channels, with stride 2 twice), mean and max pooling, a 32-dimensional placement embedding and dropout 0.3. The network on both modalities and the video-only network take the 15 video rows of a clip. The video-only probabilities are the geometric mean of two networks trained on clips cut at different offsets. All classifiers are trained with the subject-wise folds of Section 6, and we use out-of-fold predictions for every analysis.

Video feature construction. Following the WEAR feature specification, features are extracted at 30 fps from frames resized to 224×224. Row i is computed from the 16-frame input made of frame i, the 8 preceding and the 7 following frames. A one-second clip covers frames $[f_0, f_0 + 30)$, and the 15 rows supplied per test clip are those with $i \in [f_0 + 8, f_0 + 22]$, so the video side spans the same second as the inertial side. Our training and validation clips were cut differently, a choice made early in development: the graph features, the network on both modalities and one video-only network use rows $[f_0, f_0 + 14]$, and the other video-only network uses $[f_0 + 7, f_0 + 21]$. Rows before $f_0 + 8$ draw on up to eight frames (0.27 s) before the clip. Building the validation graph from the test-aligned rows changes the final macro-F1 by +0.0002 (Section 7.5). We did not retrain the classifiers on the aligned rows. Consecutive rows share 15 of 16 frames, so the 15 rows are not independent observations. For Ego-Exo4D we compute three 16-frame inputs per clip (centred on frames $f_0 + 8$, $f_0 + 15$ and $f_0 + 22$) and average them. The CLIP (`openai/clip-vit-base-patch32`, projected image embedding), DINOv2 (`facebook/dinov2-base`, pooled output) and VC-1 (`facebook/vc1-base`, class token after the final normalisation) embeddings use the single frames $f_0 + 8$, $f_0 + 15$ and $f_0 + 22$,

averaged in the same way. For the WEAR encoder comparison (Table 4) we downloaded the public WEAR videos of the training participants only. We encode the frames $f_0, f_0 + 2, \ldots, f_0 + 14$ of each clip, matching the rows that the VideoMAEv2 feature averages. Public file numbers differ from the challenge's for four recordings. We matched every video to its recording by cross-correlating the frame-to-frame change of its embedding with that of the challenge features (all at zero lag, $r = 0.74$–0.84 against at most 0.16 for the next candidate). We excluded one recording that matched only weakly ($r = 0.40$) and the two whose video we did not download.

Measuring the view on WEAR. For the analyses of Section 3 we take one frame per clip, at the centre of the feature rows the graph averages. *Scene layout and visible body*: SegFormer-B2 [8] trained on ADE20K [9], with class fractions in a 3×3 grid. The "person" pixels are almost always the wearer's own hands and legs. The scene classes are those with a mean fraction above 0.2% (18 classes). *Camera tilt*: GeoCalib [7] with its distorted-camera model. On the Aria frames of Ego-Exo4D, its gravity estimate predicts the measured head tilt with $R^2 = 0.75$ (cross-validated over takes). *Camera shake*: phase correlation between consecutive frames at 10 fps (160×90 grey). We summarise it per clip by the mean absolute shift in x and y, their standard deviations, the largest shift and the mean frame difference. On Ego-Exo4D it predicts the measured inter-frame head rotation with $R^2 = 0.89$. We subtract components by ridge regression of the per-participant centred embedding on the named quantity within each participant. *Exocentric control*: the first static camera of each Ego-Exo4D take, encoded in the same way as the egocentric video.

Hyperparameters. $k = 30$ (mutual), $\alpha = 0.75$, five iterations per stage, three stages, $\lambda = 4$, sharpening exponent 1.4, $m = 30$, $\gamma = 0.5$, $g = 0.2$, τ tuned out-of-fold per configuration (0.9 for the final system). All were selected on subject-wise cross-validation. The one-at-a-time sensitivity is reported below. The transductive stages take 0.6 s per participant in the median and at most 1.8 s (1,900–6,600 clips) on an 8-core desktop CPU, without a GPU. The point at which video enters matters. Pooling after every stage (0.8429) is slightly above pooling once after the last stage (0.8411) and no pooling (0.8301). The dose–response in g is interior-optimal, and a confidence-gated weight performs worse than the uniform one.

Sensitivity to hyperparameters. We varied one parameter at a time around the final configuration ($k \in \{15, \ldots, 90\}$, zero to three rebuilds, $\alpha \in [0.6, 0.85]$, 3–12 iterations, sharpening exponent 1.0–2.0, $g \in [0.1, 0.4]$, $m \in [20, 50]$, $\gamma \in \{0, 0.25, 1\}$). No setting improves on both seeds. The settings $k \geq 60$, $\alpha \leq 0.6$, no sharpening, $g \geq 0.4$ and no rebuilding each reduce macro-F1 by 0.006 to 0.022. The schedule is at a local optimum.

B Additional Results

Table 9 breaks the three predictors down by class. Video outperforms the inertial classifier in only five classes. Taking the better modality per class would raise macro-F1 only from 0.645 to 0.668. The gain to 0.854 comes from pooling across placements through the graph rather than from choosing a modality per class. Figure 7 shows the predictions over one participant's recording, and Figure 8 shows the effect of propagation in the video embedding.

C The Base Classifier's Contribution

Averaging the single-sensor classifier's predictions over all four sensors of the same clip (privileged, unavailable at test) raises its standalone accuracy by +0.109 but the system by only +0.014 (Figure 9). A +0.01 system gain requires about +0.077 standalone. Biasing graph edges towards cross-placement pairs has no effect either (cross-placement neighbour rate 0.75 $\rightarrow$ 0.98, macro-F1 0.8322 $\rightarrow$ 0.8312). The oracle averages four views of the same second, whereas the graph averages different seconds

Table 9. Per-class F1 of the inertial-only classifier, the video-only classifier and the final system (seed 42), with the inertial classifier's recall split by arm and leg sensors. Bold marks the stronger single modality where the gap exceeds 0.05.

	F1			inertial recall	
class	inertial	video	final	arm	leg
stretching (hamstrings)	0.57	**0.73**	0.84	0.49	0.71
lunges (complex)	0.51	**0.66**	0.81	0.65	0.36
stretching (shoulders)	0.31	**0.39**	0.80	0.47	0.05
sit-ups	0.57	0.61	0.86	0.58	0.60
jogging (rotating arms)	0.68	0.69	0.84	0.82	0.53
sit-ups (complex)	0.62	0.61	0.84	0.57	0.74
stretching (triceps)	0.49	0.47	0.94	0.76	0.10
NULL	0.76	0.73	0.88	0.79	0.76
push-ups (complex)	**0.60**	0.53	0.75	0.52	0.62
burpees	**0.67**	0.59	0.87	0.63	0.69
lunges	**0.54**	0.45	0.79	0.47	0.57
stretching (lumbar rot.)	**0.78**	0.69	0.92	0.81	0.84
push-ups	**0.65**	0.55	0.78	0.69	0.68
bench-dips	**0.62**	0.51	0.86	0.77	0.46
jogging	**0.77**	0.58	0.87	0.90	0.70
stretching (lunging)	**0.58**	0.34	0.86	0.49	0.61
jogging (sidesteps)	**0.85**	0.57	0.92	0.80	0.93
jogging (butt-kicks)	**0.84**	0.51	0.88	0.74	0.87
jogging (skipping)	**0.85**	0.43	0.92	0.80	0.91
macro-F1	0.645	0.560	**0.854**		

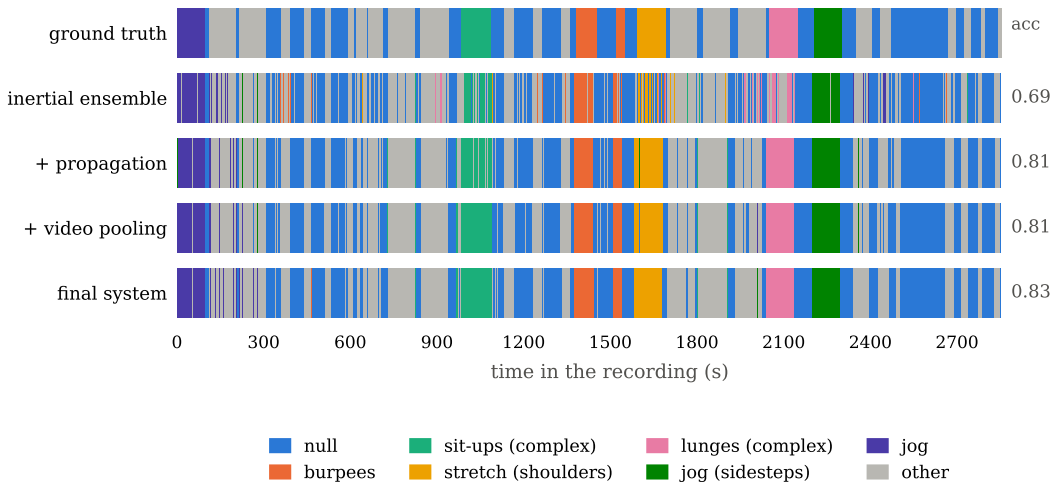


Fig. 7. Ground truth and each stage for one participant (chosen as the one whose gain from the last component is closest to the median). Temporal order is used for this figure only, and the method never uses it.

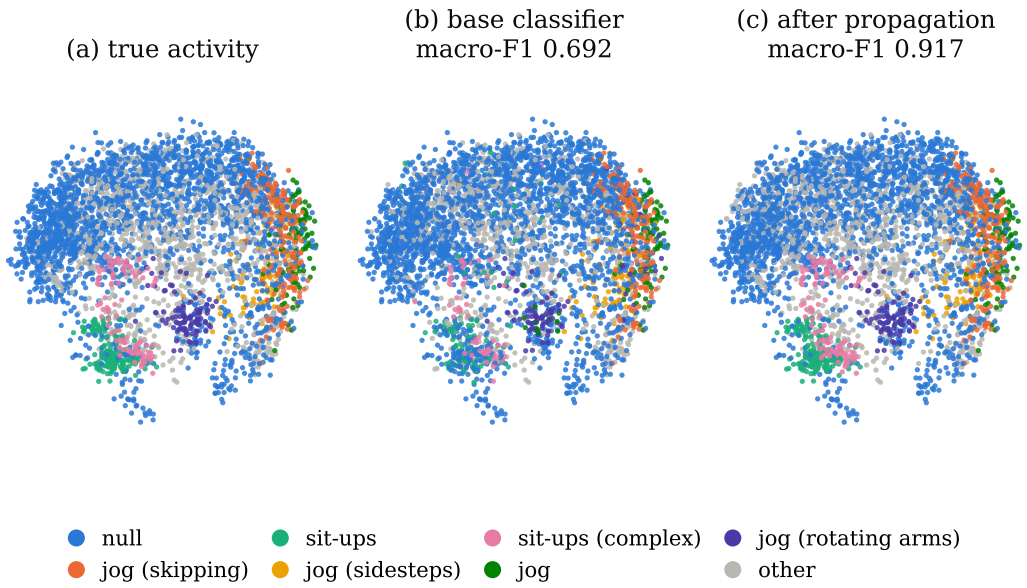


Fig. 8. Propagation on real data. Points are one participant's clips in a 2-D projection of their video embeddings, and colour is the activity. (a) ground truth; (b) the inertial classifier; (c) after propagation. macro-F1 on this participant rises from 0.692 to 0.917.

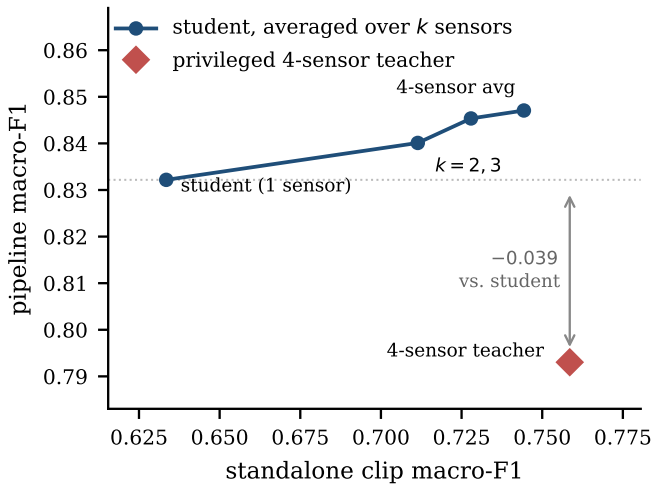


Fig. 9. At the base position, higher standalone accuracy yields little system accuracy. Averaging the student over k sensors of the same clip (blue, privileged) traces a slope of about 0.13. The four-sensor teacher (red) lies below this line.

of the same set. Of the 106 sets that an earlier configuration predicted entirely wrong, 42.5% are

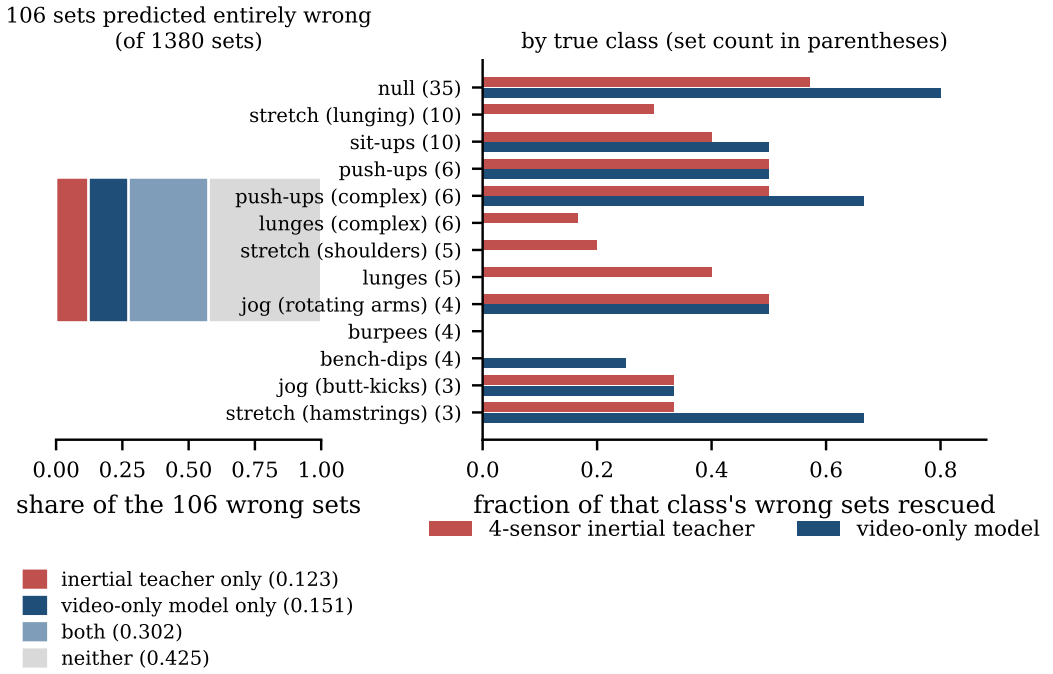


Fig. 10. Decomposition of the 106 sets predicted entirely wrong. Only 42.5% are missed by both a four-sensor inertial teacher and a video-only model, and 15.1% are recoverable by video alone. The split depends on the class, and video dominates on NULL.

missed by both a four-sensor teacher and a video-only model, and 15.1% are recoverable by video alone (Figure 10).

D Explanations Tested for the Effect of Component Removal

Look-alike clips are captured by the wrong class. The clips that removal corrects are those whose video neighbourhood belonged to another activity. Their neighbourhood purity in the raw graph is 0.42–0.54, against 0.75–0.80 overall. Of their raw errors, 67–71% (with the top 30 removed, and 75–79% with the activity means subtracted) are exactly the class that surrounded them. The captures follow one pattern in all four encoders. NULL clips are captured by the activity next to them (jogging, sidesteps, the sit-up and push-up variants, burpees, bench-dips), and static stretches are captured by NULL. Subtracting the activity means lifts the neighbourhood purity of these clips from 0.42–0.46 to 0.75–0.78. The gain arrives through re-estimation. A single stage achieves a fifth to two fifths of the improvement from three stages. Across classes, the gain is predicted by how much re-estimation reduces the edges to other activities (Spearman ρ 0.75–0.96 for removal, 0.64–0.81 for subtraction). In domain terms, a head camera records where and in which posture an activity is done, and far less of the movement that defines the activity. The seconds before and after a set, labelled NULL, are spent at the same spot in the same posture, and a static stretch looks like standing still. Their appearance therefore matches the neighbouring activity, while the limbs differ in whether they move. The appearance component pulls such clips into the look-alike class, propagation overrides their inertial evidence, and re-estimation fixes the error in place. Without it, the view still groups the clips of a session, and the inertial classifier assigns the labels.

Table 10. Graph built on video features after subtracting a limb-predicted component (two seeds). We denote by m the number of leading components removed afterwards. "Test" marks variants a test clip can supply.

Subtracted component	$m = 0$	$m = 30$	Test
none (raw features)	0.8448	0.8523	✓
four limbs, MLP	0.8502	**0.8555**	
four limbs, linear	0.8500	0.8553	
four limbs, gradient boosting	0.8491	0.8535	
four limbs, MLP, permuted	0.8372	—	
one limb, same for every clip	0.8475	0.8544	
one limb, the limb each clip wears	0.8430	0.8511	✓
one limb, same-limb pairs only[†]	—	0.8534	✓
one limb and video, factor model	0.8457	0.8534	✓
top-10 CCA directions projected out	0.8448	0.8519	✓

[†] Pairs wearing the same limb are compared on the residual, and all other pairs on the baseline features.

The limb-predicted component is unavailable at test time. For VideoMAEv2, with the video-probability block in the graph, subtracting the limb prediction gives 0.8502, against 0.8448 on the raw features and 0.8523 with removal (Table 10). Subtraction and removal are partly complementary (0.8555 together). Predicting the video better brings no further gain, since gradient boosting has the highest R^2 and the smallest gain. Projecting out a fixed subspace (the ten video directions most correlated with the limbs) has no effect. Only subtracting a per-clip amount set by that clip's limbs improves the result.

A test clip carries one limb, and the account does not transfer to it. A prediction from one limb improves the result when it is the *same* limb for every clip (0.8544), but not when each clip uses the limb it wears (0.8511). Different limbs explain different parts of the view, which distorts distances between clips wearing different limbs. Centring per placement has no effect. The missing limbs also cannot be completed well enough to restore consistency. A rank-120 factorisation reconstructs all four limbs' features almost perfectly ($R^2 = 0.96$). From one limb and the video graph, however, they are recovered only to $R^2 = 0.37$ by graph-regularised tensor completion, and to 0.41 at best. The best test-applicable variants change macro-F1 by +0.0011, which is within the seed-to-seed noise. The account therefore explains the effect of removal in part, but it offers no way to exploit it at test time.

Estimating the appearance component without the limbs does not raise the full system either (video-probability block and evidence pooling included, three seeds). An appearance subspace learned from the other participants' labels (the span of their activity means, or discriminant directions) improves macro-F1 over no removal (+0.002), against +0.008 for removing the top 30 components. It brings no further improvement on top of that removal. The appearance of an activity depends on where and facing which way a participant performed it, and it does not transfer between participants. Estimating the activity means within the test participant from the system's own predictions imports their errors (−0.001 to −0.003 with soft and −0.008 to −0.020 with hard assignments, relative to the current system). Even the ground-truth activity means raise macro-F1 by only +0.0045 over the current system, because the probability block and component removal already absorb most of the harm.

We tested twenty-three explanations for why removing the leading video components improves the graph. We rule out twenty. Two (items 8 and 12) fail with one limb but hold in part with four,

and Section 7.2 identifies what the four limbs capture. One (item 9) we cannot rule out. Truncated whitening performs comparably to removal, and our tests do not separate correcting anisotropy from removing specific content.

(1) **Removing slow appearance drift.** Correlation of the components with recording time is weak ($R^2 \leq 0.11$ for the first, 0.022 over 1–30). Median neighbour time separation rises with m (58.8 s to 89.3 s), but purity is uniformly lower at every time lag.
(2) **Improving neighbourhood purity.** Purity falls (0.73 at $m = 0$, 0.63 at 30) while the score peaks.
(3) **Improving placement invariance.** The video graph is at chance for placement (0.251) before and after.
(4) **Indirectly weakening propagation.** Lowering α at $m = 0$ only reduces macro-F1.
(5) **Indirectly reducing propagation depth.** Iterations are flat at $m = 0$.
(6) **Indirectly re-weighting the probability block of the graph features.** γ is already at its optimum.
(7) **Compensating a video–inertial offset.** Neighbour purity of the video graph against labels shifted by s seconds peaks at $s = 0$ before and after removal, with the same profile (Table 11). This tests the data as a whole. It does not detect an offset in a single session, such as the 10 s label delay of Section 9.
(8) **Reducing redundancy with the base classifier.** With predictions and a linear single-limb map, the candidate disappears. With a non-linear map from four limbs, it returns (Section 7.2).
(9) **Correcting anisotropy** (not ruled out). Centred features are strongly anisotropic, as for word vectors [47]. The first component holds 15.8% of the variance within a participant, the top 30 hold 77.2%, and the effective dimension is 19.5 of 768. Projecting onto the top 128 or 256 components and equalising their variance (truncated whitening, which discards the remaining components) gives 0.8527 and 0.8535, against 0.8516 for removal and 0.8429 for neither (three seeds). Projecting onto the same components without whitening gives 0.8415 and 0.8425, so the gain comes from equalising the variance, not from the truncation. A near-zero mean pairwise cosine (0.0015) does not guarantee isotropy.
(10) **A coarse/fine division of labour.** 80.3% of base errors cross coarse groups, and the leading components also contain within-group signal.
(11) **Reducing hubness.** Hubs [48] are more accurate than average, and explicit local scaling does not reproduce the gain.
(12) **Residualising against the inertial features.** From one limb, linear residualisation is worth −0.0007 to +0.0004. From four limbs, it recovers a large part of the gain (Section 7.2).
(13) **Visibility of the wearer's body** (external labels from a pose detector). The features do not predict it across recordings, and the fitted directions are not enriched in the top-30 subspace (0.025–0.042 against 0.039 for a random direction).
(14) **Camera egomotion** (external labels from phase correlation). Its direction occupies 0.0392 of the top-30 subspace, against 0.0391 at random.
(15) **Posture confusions shared with the classifier.** The correlation between the graph's confusions and the classifier's rises with removal (0.636 → 0.800), which is the opposite sign.
(16) **Separating different sets of the same activity.** The same-set fraction among same-label neighbours is flat (0.646–0.652).

Table 11. Neighbour purity of the video graph against labels shifted by s seconds.

	$s=-2$	$s=-1$	$s=0$	$s=+1$	$s=+2$
$m=0$	0.719	0.732	**0.732**	0.719	0.703
$m=30$	0.621	0.632	**0.633**	0.622	0.608

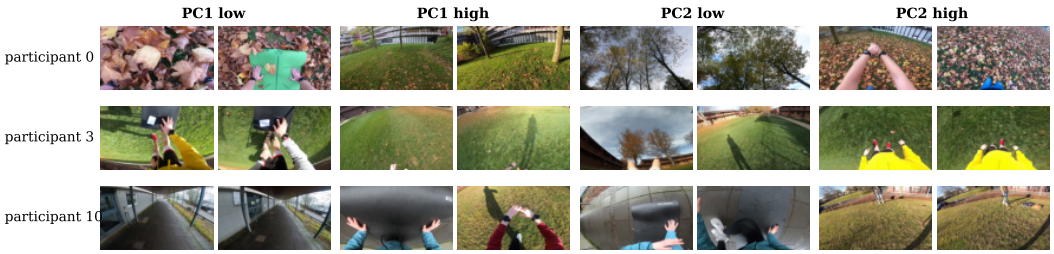


Fig. 11. Frames at the two ends of the first and second principal components of the video embedding (VideoMAEv2, one recording per participant). Frames from WEAR [1] (CC BY-NC-SA 4.0, https://creativecommons.org/licenses/by-nc-sa/4.0/), downscaled.

(17) **A contaminated subspace estimate.** Estimating the subspace from the 70% least-moving clips reduces macro-F1 by 0.0022.

(18) **Removing head orientation itself.** Projecting out the directions that measured head tilt spans on Ego-Exo4D, transferred to WEAR, changes macro-F1 by +0.0003 to +0.0011, as much as random directions. Subtracting the part explained by the camera tilt estimated on WEAR has no effect (Section 7.2).

(19) **Discarding the view.** Subtracting what the scene layout explains reduces macro-F1. Subtracting what the visible body or the camera shake explains changes nothing (Table 5).

(20) **Errors of the classifier that follow the video.** Shuffling the classifier's predictions among clips of the same participant and class retains the gain and enlarges it.

(21) **Letting the label block steer re-estimation.** Raising λ without removal lowers macro-F1.

(22) **Purity dominated by NULL.** Class-averaged purity without NULL also falls with removal (0.769 → 0.673).

(23) **Scattering concentrated wrong neighbours.** Removal lowers the share of the largest wrong class among wrong neighbours, but across classes this share is negatively related to the gain.

E Frames at the Ends of the Leading Components

Figure 11 extends Figure 1a to three participants. For each recording, we centre the embedding and compute its first two principal components. We show two frames at each end, at least 60 s apart. At every end, the frames share what the camera looks at, either the ground or the wearer's own body close up, or the scene ahead or above. For participant 10, the first component also separates two places (a covered walkway and a field).

F Other Negative Results

- **Labelled anchors.** Adding a vote from each test clip's top-30 video neighbours among labelled training clips reduces macro-F1 at every dose (-0.012 at $h = 0.1$ to -0.074 at 0.35), although the vote by itself scores 0.449.
- **Subject-adaptive self-training** on confident pseudo-labels: $0.832 \longrightarrow 0.70$–$0.75$.
- **Per-class decision weights** (19 parameters): +0.025 in cross-validation, -0.006 on the hidden test set.
- **Per-participant Sinkhorn balancing** with a uniform non-NULL target: +0.017 in cross-validation and -0.038 on the hidden test set. With a per-participant NULL target, there is no loss and no gain.
- **Metric learning** on the video features (within-recording contrastive), hard clustering, confidence-weighted propagation and per-recording video standardisation: equal or worse.
- **Temporal order recovery from video**: successor top-1 ≤ 0.35 by three methods.
- **Cascaded propagation of a stacked classifier's output** and **graph ensembles**: about +0.002, below the decidable threshold.
- **Mirror-canonicalised inertial features**, mapping left limbs onto the right-limb frame by sign flips chosen on training data: +0.005 for the inertial classifier on its own, and +0.0002 to +0.0006 in the full system (three seeds). Propagation largely absorbs improvements of the base classifier.

G Adjacency of Shuffled Clips

When two consecutive clips wear the same limb, the last sample of one and the first of the next are 20 ms apart. On training participants cut the same way, the gap between them (median 0.052 g) is the size of one within-clip step (0.047 g), against 1.25 g for an arbitrary same-limb pair. Consider the clips whose successor wears the same limb. For 0.681 of them, ranking same-limb candidates by the extrapolated gap finds the true successor first, and adding video continuity raises this to 0.750. Among the most confident fifth, 99.2% are correct. The signal is weaker than these figures suggest, because three quarters of successors wear a different limb, where only video continuity is available (top-1 0.26). It is also weaker because near-static clips match by chance (best-candidate gap: median 0.023 g for true successors, 0.065 g otherwise). A learned link scorer with a linear assignment links half the clips with label agreement 0.95. The smoothing of Section 8.2 is measured along these links. The adjacency was measured on training participants and on the gap distribution of the test set, and it is used in no reported result.

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